

EEG Datasets

BioRTC 2026 - Yobe State University

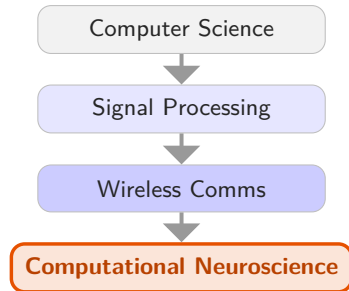
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June 2026

The journey



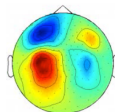
Following questions



What do I work on?

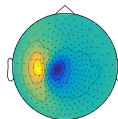
EEG

Oscillations, connectivity & time-frequency analysis of language and cognition



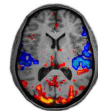
MEG

Source-level spatiotemporal dynamics; high-density sensor arrays



fMRI

Language network mapping; BOLD mediation modelling



My main focus: Non-Invasive Brain Imaging & Language

What Data Will We Work With?

TUH EEG Corpus (Temple University)

Three public sub-corpora from a single clinical EEG bank:

- **TUAR** v3.0.1 *Artifact annotations*
- **TUSZ** v2.0.6 *Seizure labels*

Local Clinical Recording

SUBJECT1.edf

Yobe State University Teaching Hospital (YSUTH) - real patient, no ground-truth labels

Session	Dataset	Purpose
Workshop (Fri)	TUAR	Preprocessing demo
Workshop (Fri)	TUAR	CNN seizure detection
Project track 1	TUSZ	DL classification
Project track 2	TUSZ	Source reconstruction
Both projects	YSUTH EDF	Transfer / inference

The TUH EEG Corpus

- Collected at **Temple University Hospital** (Philadelphia, USA)
- Standard clinical EEG: scalp electrodes, 10-20 system
- One of the **largest public EEG repositories** - tens of thousands of sessions
- Recordings span routine clinical EEGs from adult patients, many with epilepsy

Corpus	Recordings	Labels
TUAR v3.0.1	310	Artifact types
TUSZ v2.0.6	~6,000	Seizure on/off
TUEP v3.1.0	200	Epilepsy / control

Important

All three share the same EDF format and referential storage - pipelines transfer across them.

Data Access and Use Policy

For this course

The TUH EEG data is shared under a **Data Use Agreement (DUA)** signed by me covering all participants.

- Data has been curated and distributed for teaching use only
- **You must delete all data after the course ends**
- Do not share, publish, or redistribute any recordings
- Post-course research use requires your own registration

Download it yourself

Register at the NEDC portal - free, academic use:

https://isip.piconepress.com/projects/tuh_eeg/

Once approved you can access TUAR, TUSZ, TUEP and the full corpus via rsync.

Course materials

Notebooks and slides:

https://github.com/comptelab/BIORTC_Nigeria

TUAR - Artifact Corpus

What it is: 310 clinical EEG recordings with time-stamped artifact annotations.

Annotation format: one .csv per recording channel, start_time, stop_time, label, confidence

Artifact types annotated:

- elec - electrode artifact (bad contact, pop)
- eyem - eye movement / blink
- musc - muscle (EMG contamination)
- eyem_musc - mixed eye + muscle

Demo file we will use

aaaaabbn_s010_t000.edf

305 s, 19 channels

Events in first 120 s:

- 3-36 s: elec on C3-CZ
- 36-40 s: eyem (frontal)
- 91-112 s: eyem_musc (frontal)

Why it matters

Real artifact labels let us ask: *can ICA remove all of these?* (Spoiler: not always.)

What it is: Benchmark for automated seizure detection - continuous clinical EEGs with expert-annotated seizure events.

What you will do with it:

1. Train a 1D-CNN to detect seizure windows
2. Make it work *online* (no future data)
3. Apply it to an *unlabeled* local recording

Seizure demo file

aaaaakbz_s001_t000.edf

1442 s (≈ 24 min), 19 channels

Two seizures (complex partial):

- Seizure 1 at ≈ 439 s
- Seizure 2 at ≈ 940 s

Key challenge

Seizures are **rare events** ($< 5\%$ of recording time). Class imbalance is not just a stats problem - it changes everything about how you build and evaluate the model.

The Local Data - YSUTH Patient EEG

Where it comes from: Yobe State University Teaching Hospital (YSUTH) - 18-year-old patient, no ground-truth labels.

Recording:

- 19 EEG channels (10-20) + A1/A2 + auxiliary (SpO2, EtCO2, pulse...)
- **500 Hz**, 24.2 minutes continuous

Annotations during the session:

- Eyes open/closed; montage changes
- Photoc stimulation; hyperventilation (90 s)
- No seizure labels - clinical read is PDF only (not machine-readable)

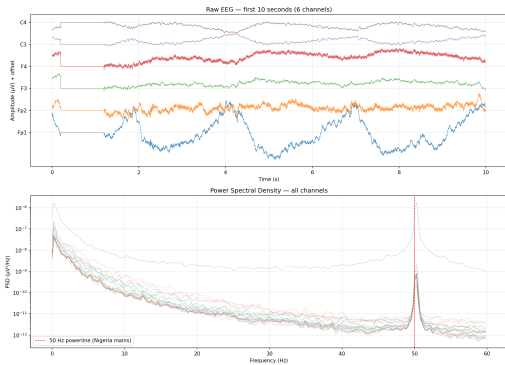
The real challenge

Your model is trained on labeled TUH data. Can it generalize to a different hospital, equipment, and sampling rate?

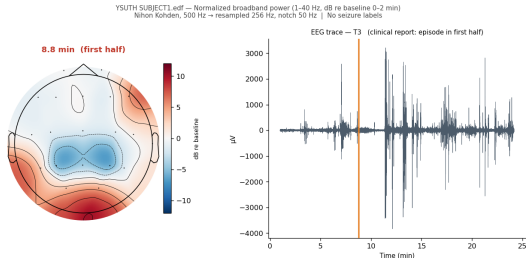
This is the **domain adaptation** problem in clinical AI.

Note: The recording is anonymized.

The Local Data - What It Looks Like



Raw trace + PSD



Normalized power topomap, full 24-min recording at
16 $\times$ speed

[Click to play \(yobe_topomap.mp4\)](#)